Chemistry - Undergraduate Level Assessment 15 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The solid-state structures of the principal allotropes of elemental boron are made up of which of the following structural units?
   (a) B_{12} icosahedra
   (b) B_8 cubes
   (c) B_6 octahedra
   (d) B_4 tetrahedra

2. What is the orbital angular momentum quantum number, l, of the electron that is most easily removed when ground-state aluminum is ionized?
   (a) 3
   (b) 2
   (c) 1
   (d) 0

3. Infrared (IR) spectroscopy is useful for determining certain aspects of the structure of organic molecules because
   (a) all molecular bonds absorb IR radiation
   (b) IR peak intensities are related to molecular mass
   (c) most organic functional groups absorb in a characteristic region of the IR spectrum
   (d) each element absorbs at a characteristic wavelength

4. Reduction of D-xylose with NaBH 4 yields a product that is a
   (a) racemic mixture
   (b) single pure enantiomer
   (c) mixture of two diastereomers in equal amounts
   (d) meso compound

5. Which of the following is a primary standard for use in standardizing bases?
   (a) Ammonium hydroxide
   (b) Sulfuric acid
   (c) Acetic acid
   (d) Potassium hydrogen phthalate

True / False (10 marks)

Answer True or False

6. True or False: At standard temperature and pressure, Br_2 is a colorless gas.
7. True or False: SCl 2 is least likely to behave as a Lewis acid due to its lack of electron-deficient central atom.
8. True or False: Protons and neutrons possess orbital angular momentum only.
9. True or False: Tetramethylsilane's protons are quite strongly shielded, which is why it is used as a reference in NMR spectroscopy.
10. True or False: The infrared-active normal modes of a carbon dioxide molecule include bending and asymmetric stretching.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the factors that contribute to the limiting high-temperature molar heat capacity at constant volume (C_V) of a gas-phase diatomic molecule, including its theoretical derivation and implications.
12. Discuss the implications of the Heisenberg uncertainty principle for a quantum mechanical particle in the lowest energy level of a one-dimensional box, focusing on the limitations of measuring position and momentum.

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